CS471 Status Report: Hardware Decompression for Energy-Efficient Local LLMs

Mihai Indreias Filip Corlan
EPFL, Lausanne, Switzerland

I. Response to Proposal Feedback

A. Share of linear layers in GPU time

> "Your hypothesis is that linear layers are 30% of GPU execution time, but you do not provide any insights to support this claim."

The 30% figure in the proposal was a rough estimate, and we have now measured it. We ran our target model, Qwen3.8-27B in the quantized version UD-Q4_K_M, on an NVIDIA H100 GPU and recorded how long the GPU spends on each kind of operation. An LLM works in two phases: it first reads the user's prompt (prefill), and then generates the answer one token at a time (decode). Matrix multiplications with the model's weights take 84% of GPU time in decode and 76% in prefill, so the hypothesis holds with a large margin.

B. Choice of numerical formats

> "Half this report is explaining the numerical formats and lacks discussion on why these formats were chosen... Is it necessary to evaluate all these formats? Can preliminary experiments identify similar architectural implications, so that findings generalize across groups of formats?"

We study only the formats used by our target model (Table 1), which is distributed in GGUF, the format that most people use to run LLMs on their own machines. These are therefore the formats that matter in practice for local inference, which processes one request at a time and is limited by energy, exactly the setting we want to study.

It is not necessary to study every format separately. Our first measurements show that the formats fall into three groups, according to the extra work needed to decompress them rather than according to their size: formats with a single scale per block, formats with two levels of scales and offsets (the "K" formats), and formats that look up values in a small table (the "IQ" formats). Formats in the same group behave similarly, so we reason about these three groups and expect our findings to carry over to other formats in each group.

For background, GGUF is the file format of llama.cpp, the engine behind most tools for running LLMs locally. A GGUF file stores the weights of a model in compressed form. Each weight matrix is split into small blocks of 32 or 256 weights. Every weight in a block is stored as a small integer of 2 to 8 bits, and the block also stores a shared scale factor (and in some formats an offset) that converts these integers back into real values. The formats differ in how many bits they use and in how the scales are organized, and a single model usually mixes several of them, using more bits for the weights that matter most for accuracy.

II. Update on Problem Definition

The problem we study has changed since the proposal. The proposal asked whether hardware support for compressed formats, as in the Avant-Garde design [1], makes LLMs run faster. Our measurements show that speed is the wrong goal for local inference. To generate each token, the GPU must read every weight of the model from memory once (15.5 GB for our model), so the speed is limited by how fast memory can deliver data, not by how fast the GPU can compute [2, 6]. Faster arithmetic would not help.

We therefore look at energy instead. Compressing the weights reduces the amount of data read from memory, and reading data from memory costs far more energy than computing on it [3]. However, no GPU can read these formats directly, so the GPU has to decompress every weight in software before using it, and this costs energy too. We found that this cost is large: the most common format makes the weights 3.6 times smaller than uncompressed 16-bit weights, but reduces energy by only 2.2 times. Our new question is how much energy is spent on this software decompression, and how much of it a small dedicated hardware unit could save.

Table 1: The weight formats used by our model, the group each belongs to, and their share of the data read and of the GPU time spent while generating text.

Format	Group	Share of data	Share of time
IQ4_XS	Lookup table	37%	29%
Q5_K	Two-level	34%	26%
Q4_K	Two-level	16%	14%
Q6_K	Two-level	9%	6%
IQ4_NL	Lookup table	3%	2%
Q8_0	Single scale	0.5%	5%

III. Update on Design Space

A. New dimensions

Our new focus adds five dimensions to the design space. Table 2 summarizes them by comparing the choices made by Avant-Garde [1] with ours.

Table 2: New design dimensions, comparing Avant-Garde with our approach.

Dimension	Avant-Garde	This project
Part of the GPU	Tensor cores	General-purpose cores
Where weights are decompressed	Expanded in memory	Compressed until just before use
Where the inner scale is applied	Folded into every weight	Once to each sub-block's sum
Formats	Multi-level power-of-two scales	Also non-power-of-two scales, offsets, lookup tables
Metric	Speed	Energy, total and above idle

These choices follow from our setting. When generating text for a single user, the GPU uses its general-purpose cores rather than the tensor cores, the specialized matrix units that Avant-Garde extends. Storing weights in expanded form roughly doubles the data read from memory, which is exactly the cost we want to avoid, so we keep them compressed until the last moment. Both Avant-Garde and GGUF use multi-level formats, in which a large block has one scale and each smaller sub-block inside it has a second, inner scale. Avant-Garde handles this by folding the inner scale into every weight, which turns the format into a simpler one with a single scale per block but makes every weight wider. We instead first multiply and add up the weights of a sub-block as small integers, and then multiply the result by the inner scale once. This needs one multiplication per sub-block instead of one per weight, and keeps the arithmetic narrow. Avant-Garde also only supports scales that are powers of two, whereas GGUF uses arbitrary scales, offsets and lookup tables.

B. New related work

Avant-Garde [1] adds hardware support for multi-level formats with power-of-two scales to a simulated GPU, but does not cover the other features of GGUF formats or the single-user setting. Other recent work designs hardware for similar block formats [4] or for formats based on lookup tables [5], or speeds up lookup-table formats in software on existing GPUs [6], and we will build on these.

IV. Questions, Hypotheses and Tests

A. Status of the proposal's questions

We have answered the question of what limits LLM inference on the GPU: generating text is limited by reading weights from memory, and matrix multiplications take most of the time, which confirms our original hypothesis. We have partly answered how much the compressed formats cost, by measuring their energy on a single matrix and counting the instructions the GPU executes for each of them. The question of which hardware changes reduce this cost will be answered by our hardware design study. We have dropped the plan to simulate Avant-Garde, because the available GPU power simulator is not accurate for modern GPUs, and we no longer use speed as our main metric.

B. New hypotheses

- H1: Compressed formats cost more energy than the data they read would explain, and this extra energy, between 15% and 40%, comes from decompressing the weights in software.
- H2: A hardware decompression unit costs only a small fraction of the energy of the arithmetic it feeds, so formats that are complex for software are cheap for hardware.
- H3: Adding such a unit would reduce the energy of generating text by more than 10%.

C. Methodology

Our work has three parts. First, we measure energy on a real GPU. We run the matrix multiplications of each format on the H100 while reading its built-in energy meter, and we compare formats that read the same amount of data but need different amounts of decompression work. To find out where the extra energy goes, we count the instructions the GPU executes for each format, and we run a simplified version of the multiplication in which individual steps of the decompression can be switched off. We then measure the energy of generating text with the full model. Second, we design the hardware. We describe a small arithmetic unit in a hardware description language, starting from a basic version without decompression and adding support for each group of formats in turn. We then use open-source chip design tools to estimate its area and power. Third, we combine the two. Because our design and the H100 use different manufacturing technologies, we cannot compare their energy directly. Instead, we compare the extra energy each one spends on decompression, relative to its own baseline.

Table 3: How much smaller each format makes the weights, and how much it actually reduces energy, compared with uncompressed 16-bit weights (F16). Measured on the H100 for one matrix multiplication.

Format	Group	Data reduction	Energy reduction
F16	Uncompressed	1.00×	1.00×
Q8_0	Single scale	1.88×	1.65×
Q4_1	Single scale	3.20×	2.69×
Q4_0	Single scale	3.56×	2.65×
IQ4_NL	Lookup table	3.56×	2.37×
IQ4_XS	Lookup table	3.76×	2.52×
Q6_K	Two-level	2.44×	1.72×
Q5_K	Two-level	2.91×	1.89×
Q4_K	Two-level	3.56×	2.21×

V. Results

A. Where the time goes

Our model generates about 77 tokens per second on the H100, and most of this time is spent multiplying the compressed weights with the model's input. Since every token reads all 15.5 GB of weights, the GPU reads about 1.2 TB of weights per second on average. Uncompressed weights can be read at about 2.2 TB/s, but the compressed formats of our model reach only 60–80% of this speed, because the GPU spends time decompressing them.

B. Energy per format

To measure the cost of each format, we store the same large matrix in every format and repeatedly multiply it by a vector on the H100, which is the main operation performed while generating text. During the run we read the GPU's built-in energy meter and subtract the power the GPU draws when idle, so that only the energy spent on the work itself is counted. Fig. 1 shows two quantities for each format. The top panel shows the energy spent per byte of weights read. If memory traffic were the only cost, every format would sit at the same height as uncompressed F16, because each byte would cost the same to fetch. Anything above the F16 line is therefore extra energy spent on handling the format. The bottom panel shows the speed, that is, how many bytes of weights the GPU gets through per second. F16 needs no decompression and runs close to the maximum speed of the memory. When a format runs more slowly, the memory is not the bottleneck any more: the GPU is busy decompressing.

The figure and Table 3 lead to three observations. First, compression saves less energy than it saves data. Every compressed format spends more energy per byte than F16, up to two thirds more, so a format that makes the weights 3.6 times smaller re-

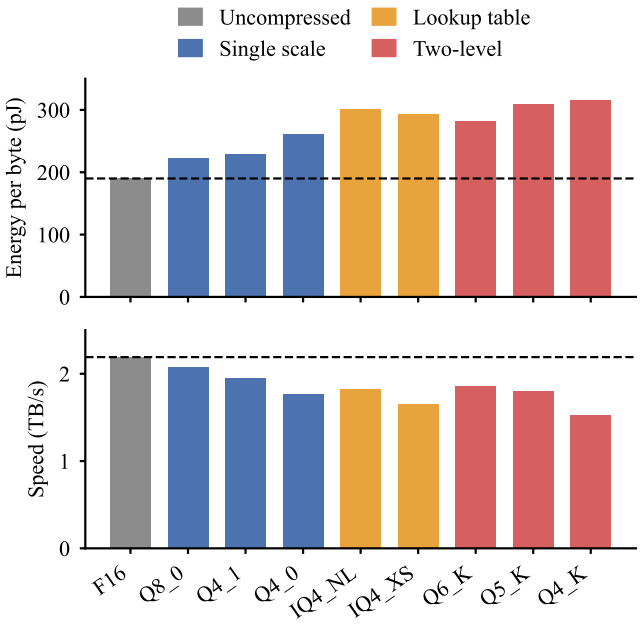


Figure 1: Energy per byte (top) and speed (bottom) when multiplying a matrix stored in each format by a vector on the H100. The dashed lines mark uncompressed F16.

duces energy by only 2.2 times. Second, the extra cost grows with the complexity of the format. Q4_0, Q4_K and IQ4_NL all store exactly the same number of bytes per weight, yet Q4_K uses 21% more energy than the simple Q4_0. Since the data read is identical, this difference must come from the work of decompressing, which supports H1. Third, the extra cost has two different causes. The two-level formats, such as Q4_K, are slower, so the GPU stays busy for longer for the same amount of data. The lookup-table formats are about as fast as the simple formats but draw more power while they run.

If every format cost as little energy per byte as F16, the energy of these matrix multiplications would drop by up to about a third. This is an optimistic upper bound. A more realistic target, bringing all formats down to the cost of the simplest compressed format, would save 15–20%.

C. *Where the extra energy comes from*

The measurements above show that complex formats cost more energy, but not why. To find out, we counted the instructions that the GPU executes for each format while multiplying the same matrix. Fig. 2 compares this count, per byte of weights, with the energy per byte from Fig. 1. The two are closely related: the formats lie almost on a straight line, with a correlation of 0.98. The line starts at about 125 pJ per byte, which we can read as the cost of moving the data itself, and every additional instruction per byte adds about 15 pJ. Under this reading, about 60% of the energy that Q4_K uses above idle goes into executing instructions rather than into reading memory. The same line also predicts the gap between Q4_K and Q4_0 well: Q4_K executes 2.3 more instructions per weight, which corresponds to 36 pJ, close to the 31 pJ we measured. F16 executes the most instructions per weight, but since each of its weights takes two bytes, it executes the fewest per byte. This is strong evidence that the extra energy is spent on computation, which supports H1, although a correlation over nine formats does not prove it.

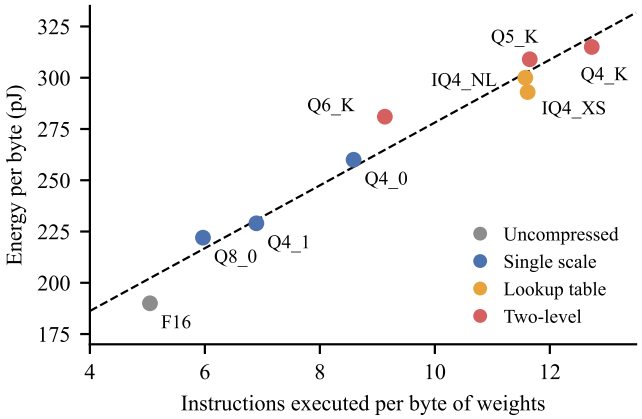


Figure 2: Instructions executed and energy used per byte of weights for each format, for the same matrix multiplication as in Fig. 1. The dashed line is a straight-line fit.

Looking at the kinds of instructions shows where this extra work goes. The instructions that unpack the bits of the weights roughly double, from about one per weight for Q4_0 to about two for the two-level and lookup-table formats. The two-level formats also need twice as many multiply-and-add instructions, because their offsets require a second sum over the input. Finally, Q4_0 needs more memory reads than Q4_1 even though it stores fewer bytes, because its 18-byte blocks are not aligned to 4 bytes and have to be read in smaller pieces. This explains why Q4_1 is cheaper than Q4_0 in Fig. 1.

To separate the steps of decompression, we also wrote a simplified version of the multiplication in which individual steps can be switched off. The results are then deliberately wrong, but the change in running time shows how much each step costs. For Q4_0, switching off decompression entirely saves only 4% of the time. For Q4_K, switching off the scales and offsets removes the whole overhead and saves 31%. For IQ4_XS, most of the overhead comes from the table lookup, and switching it off saves 44%. Our simplified version is slower than llama.cpp, so these percentages overstate the real costs, but the picture is clear: each group of formats has one dominant cost, the scales and offsets for the two-level formats and the table lookup for the lookup-table formats. These are exactly the steps that a hardware decoder would take over.

VI. Challenges

The SCITAS cluster does not allow the detailed profiling tools that would show exactly what the GPU does while decompressing. We can count the instructions it executes, but not measure how long each step takes, so we have to infer the cost of each step indirectly. The GPU's energy meter covers the whole board, including memory, so we cannot separate memory energy from computation energy. Our hardware design and the H100 are built in different technologies, so we can only compare relative costs.

VII. Update on the Plan

A. *Team composition and priorities*

The team composition is unchanged. We will first finish the GPU measurements, then design the hardware unit, and finally combine both into an estimate of the energy saved. We have

dropped the simulation of Avant-Garde.

B. Milestones

Table 4 lists our milestones.

Table 4: Planned milestones, with the week counted from the submission of this report.

Week	Milestone
1	Energy measurements for all matrices of the model, for the simplified decompression tests and for generating text with the full model
2	Hardware unit for the single-scale and two-level formats
3	Support for lookup-table formats; area and power estimates
4	Energy comparison, final estimate and report

C. Final goal

In the proposal, our goal was to compare how fast software and Avant-Garde handle compressed formats, using a simulator. Our new goal has two parts. First, we want to measure how much energy a real GPU spends on decompressing weights in software when running a local LLM. Second, we want to design a small hardware unit that extends Avant-Garde to the formats used by GGUF and decompresses weights just before they are used, and to estimate how much energy it saves per generated token and how much chip area and power it costs.

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